

Histopathology of Sweet Syndrome

Although the dermal inflammation is usually dense and diffuse, it can also be perivascular or show "secondary" changes of leukocytoclastic vasculitis, which are considered epiphenomena and not indicative of a primary vasculitis. Neutrophilic spongiotic vesicles or subcorneal pustules result from exocytosis of neutrophils into the epidermis. When neutrophils are located entirely or partially within the subcutaneous fat, the condition is termed subcutaneous Sweet syndrome.

Other Laboratory Tests

Peripheral leukocytosis with neutrophilia and an elevated erythrocyte sedimentation rate are the most consistent laboratory findings in Sweet syndrome. However, leukocytosis may be absent in some patients with biopsy-confirmed disease. In cases of malignancy-associated Sweet syndrome, findings such as anemia, neutropenia, or abnormal platelet counts may occur.

A complete blood cell count with leukocyte differential and platelet count, assessment of acute phase reactants (such as erythrocyte sedimentation rate or C-reactive protein), serum chemistries (to evaluate liver and kidney function), and urinalysis should be performed. Thyroid function testing is also recommended due to a well-documented association between thyroid disease and Sweet syndrome.

Special Tests

Evaluation for Extracutaneous Manifestations

Extracutaneous involvement in Sweet syndrome may lead to additional laboratory or imaging abnormalities. Patients with central nervous system involvement may show abnormalities on brain single photon emission computed

tomography (SPECT), computed tomography (CT), electroencephalography (EEG), magnetic resonance imaging (MRI), or cerebrospinal fluid analysis.

Those with renal or hepatic involvement may exhibit hematuria, proteinuria, or elevated hepatic enzymes. Pulmonary involvement may present with pleural effusions and culture-negative, corticosteroid-responsive pulmonary infiltrates on chest radiography.

Malignancy Workup

Given the association between Sweet syndrome and underlying malignancy—particularly hematologic and solid tumors—a thorough malignancy evaluation should be considered, especially in atypical, recurrent, or refractory cases.

Differential Diagnosis

Clinical Differential Diagnosis (Box 31-1)

The skin and mucosal lesions of Sweet syndrome can resemble those of other conditions, necessitating careful clinical and histopathologic differentiation.